Lecture 03

The Perceptron

And introduction to single-layer neural networks

STAT 479: Deep Learning, Spring 2019

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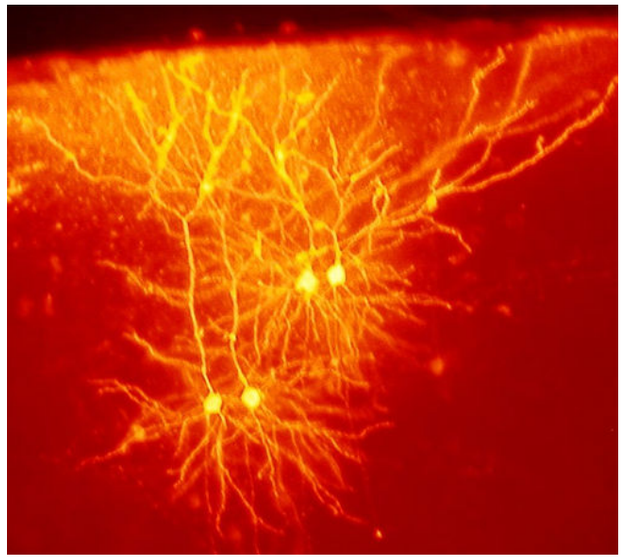
http://stat.wisc.edu/~sraschka/teaching/stat479-ss2019/

Announcements

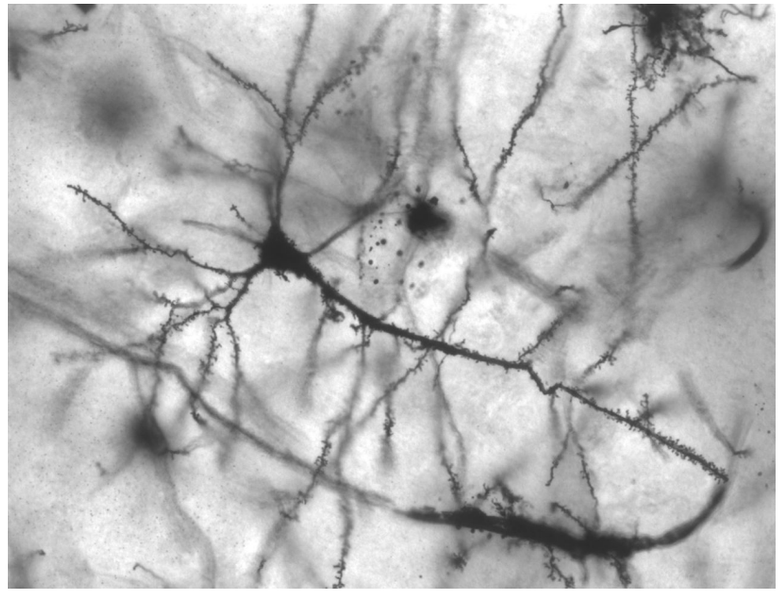
- Project groups (by next Fri), 3 members per group
- Project topics (brainstorm with group members)
- Piazza for questions, encouraged to help each other (but don't share your HW solutions)
- HW1 (by next Thu, more about that later)

Inspired by Biological Brains and Neurons

https://publicdomainpictures.net/en/view-image.php?image=130359&picture=human-brain



https://commons.wikimedia.org/wiki/Neuron#/media/File:Mouse_cingulate_cortex_neurons.jpg



https://commons.wikimedia.org/wiki/Neuron#/media/File:Pyramidal_hippocampal_neuron_40x.jpg

Number of neurons in brains ...

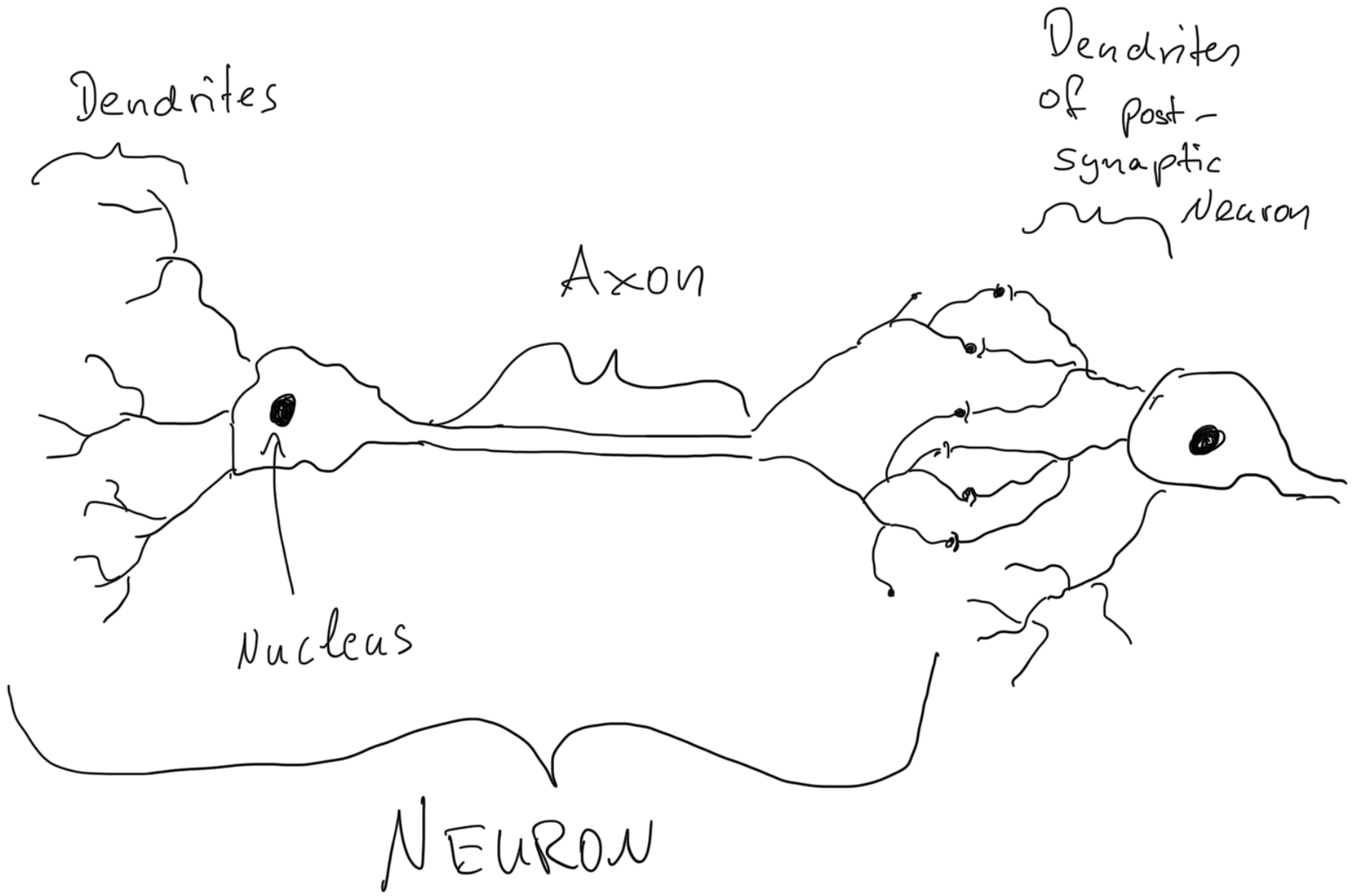
Name	Neurons in the brain/whole nervous system	Synapses	Details	Image	Source
Sponge	0				[3]
Trichoplax	0				[4]
Ciona intestinalis larva (sea squirt)	231	8617 (central nervous system only)			[5] [6]
Asplanchna brightwellii (rotifer)	about 200		Brain only		[7]
Caenorhabditis elegans (roundworm)	302	~7,500			[8]
Jellyfish	5,600		*Hydra vulgaris* (*H. attenuate*)		[9]
Fin whale	15,000,000,000		*Balaenoptera physalus*		[55]
Human	16,000,000,000		*Homo sapiens*: (For average adult) "The human cerebral cortex, with an average 1233 g and 16 billion neurons, is slightly below expectations for a primate brain of 1.5 kg, while the human cerebellum, at 154 g and 69 billion neurons, matches or even slightly exceeds the expected"		[43][2][56]
Long-finned pilot whale	37,200,000,000		*Globicephala melas*: "For the first time, we show that a species of dolphin has more neocortical neurons than any mammal studied to date including humans."		[57]

Source: https://en.wikipedia.org/wiki/List_of_animals_by_number_of_neurons

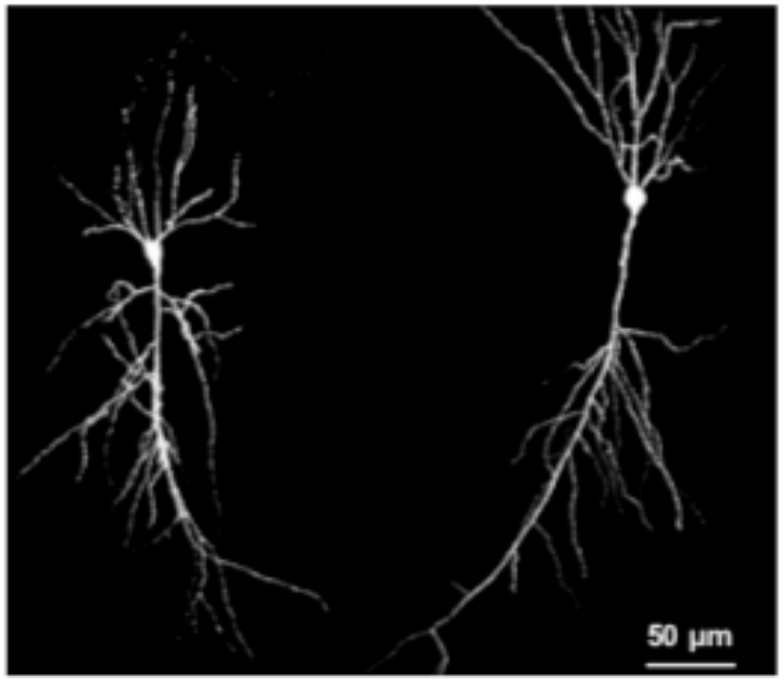
On a sidenote:

Name	Short scale (US, Eastern Europe, English Canadian, Australian, and modern British)	Long scale (Western, Central Europe, older British, and French Canadian)
Million	10^6	10^6
Milliard		10^9
Billion	10^9	10^{12}
Billiard		10^{15}
Trillion	10^{12}	10^{18}

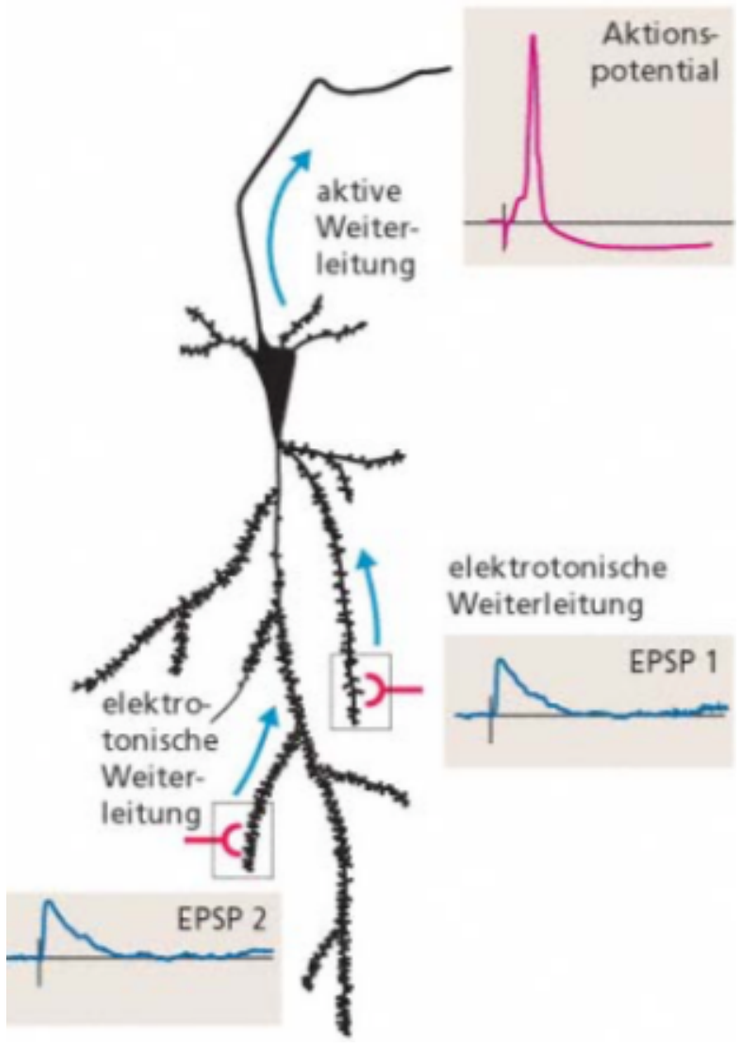
A Biological Neuron



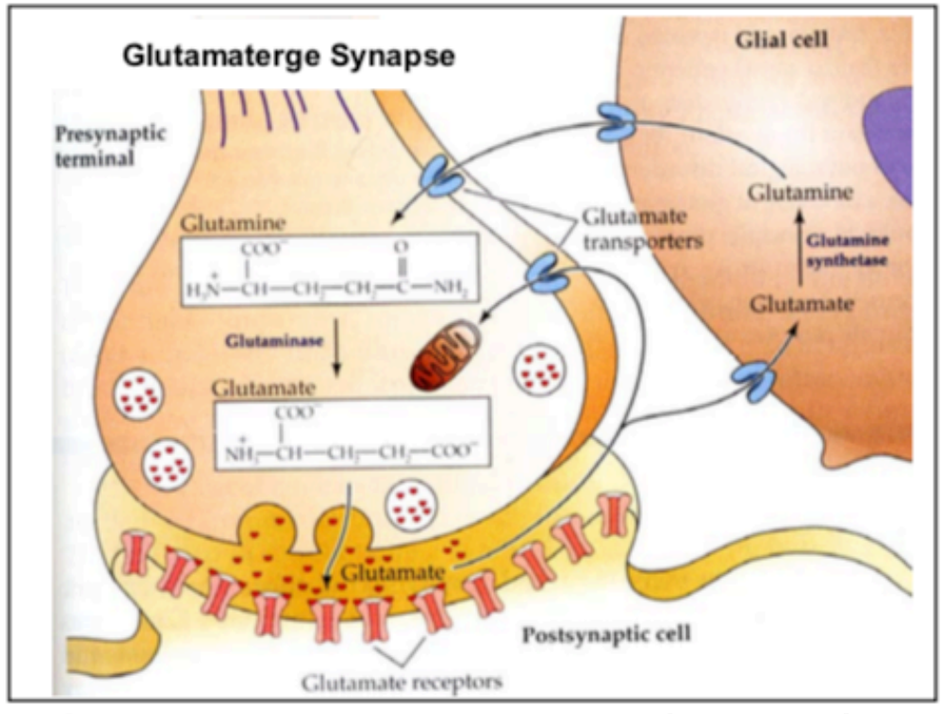
Biological Neurons



Pyramidal neuron cells in mouse cortex



electrical postsynaptic potential accumulates; when it reaches a threshold => action potential signal

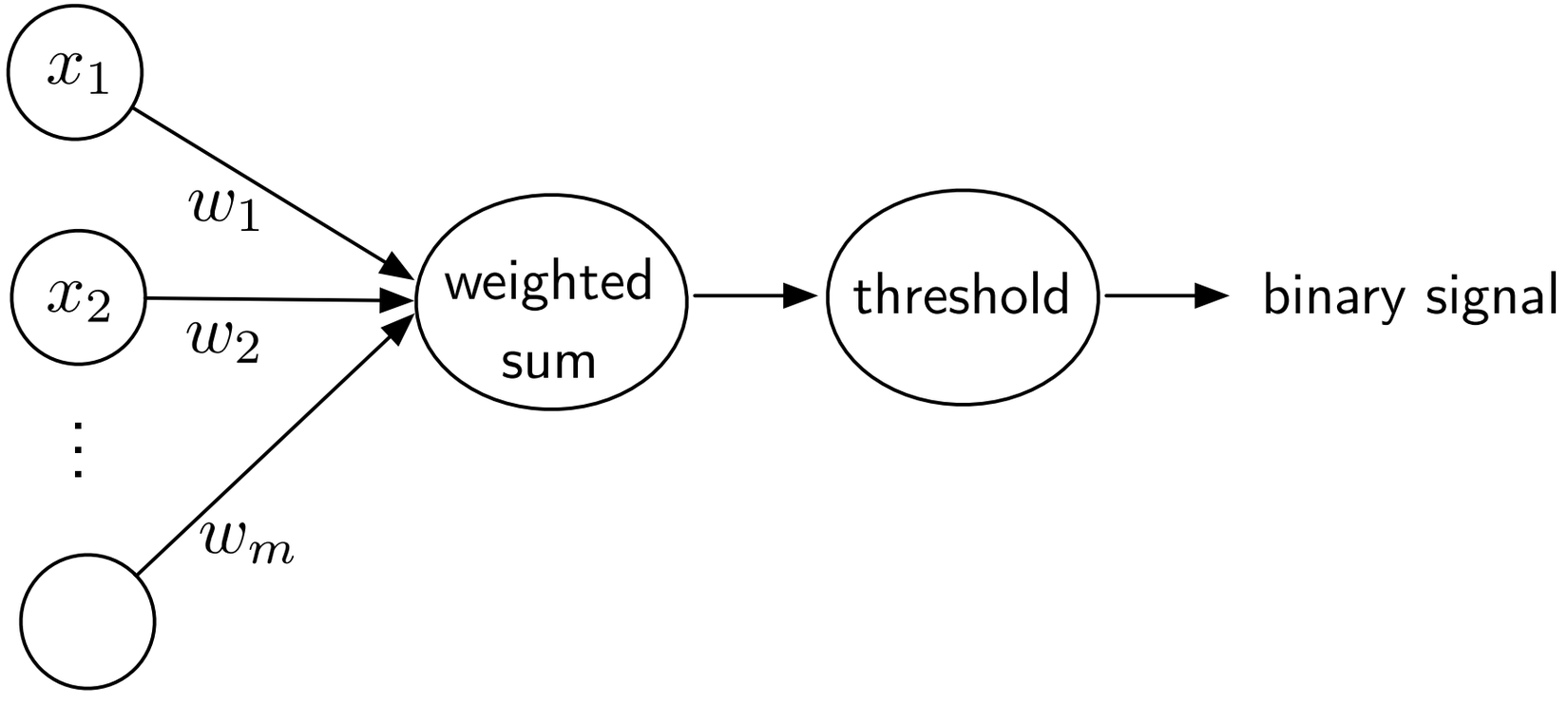


Synaptic connection is chemical

McCulloch & Pitts Neuron Model

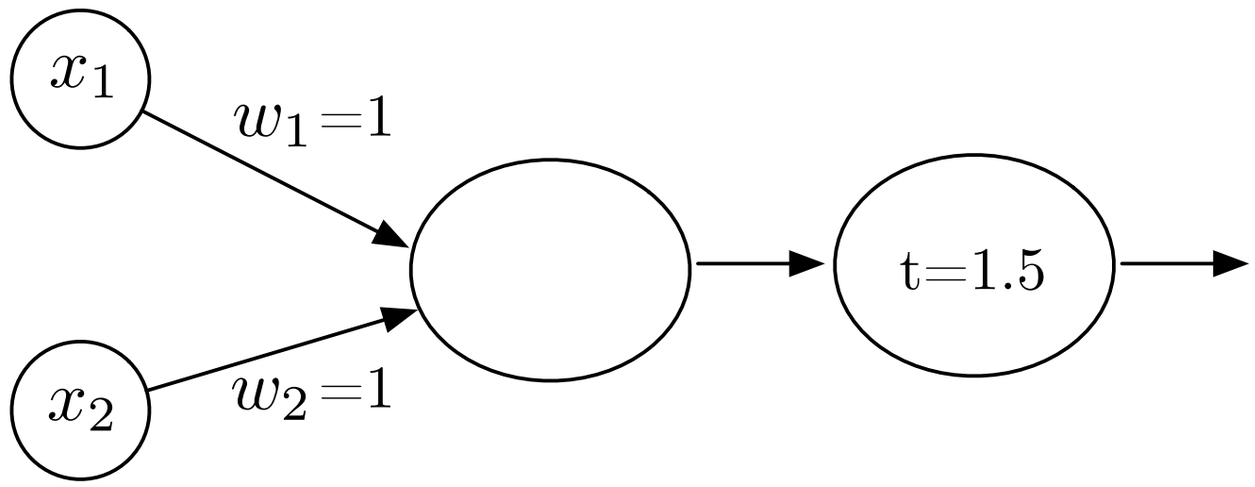
A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH and WALTER H. PITTS 1943



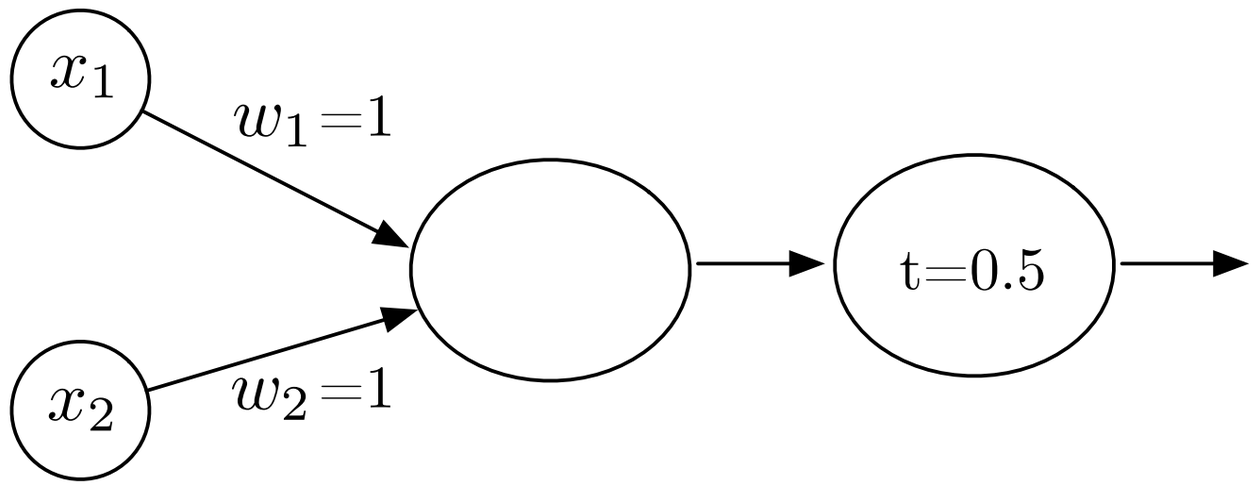
Logical AND Gate

x_1	x_2	Out
0	0	0
0	1	0
1	0	0
1	1	1



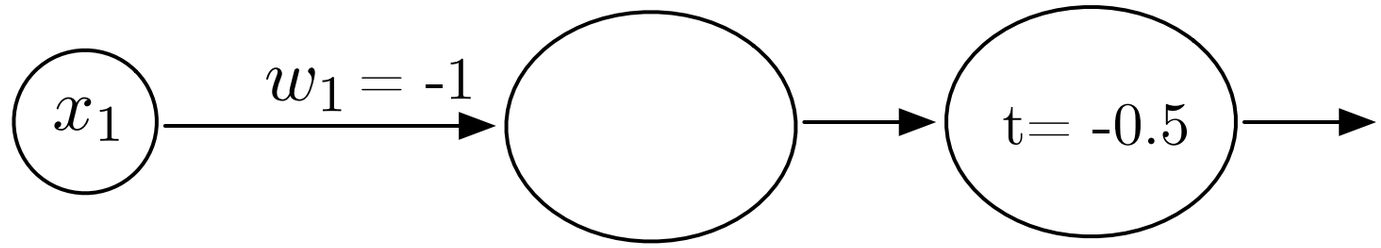
Logical OR Gate

x_1	x_2	Out
0	0	0
0	1	1
1	0	1
1	1	1



Logical NOT Gate

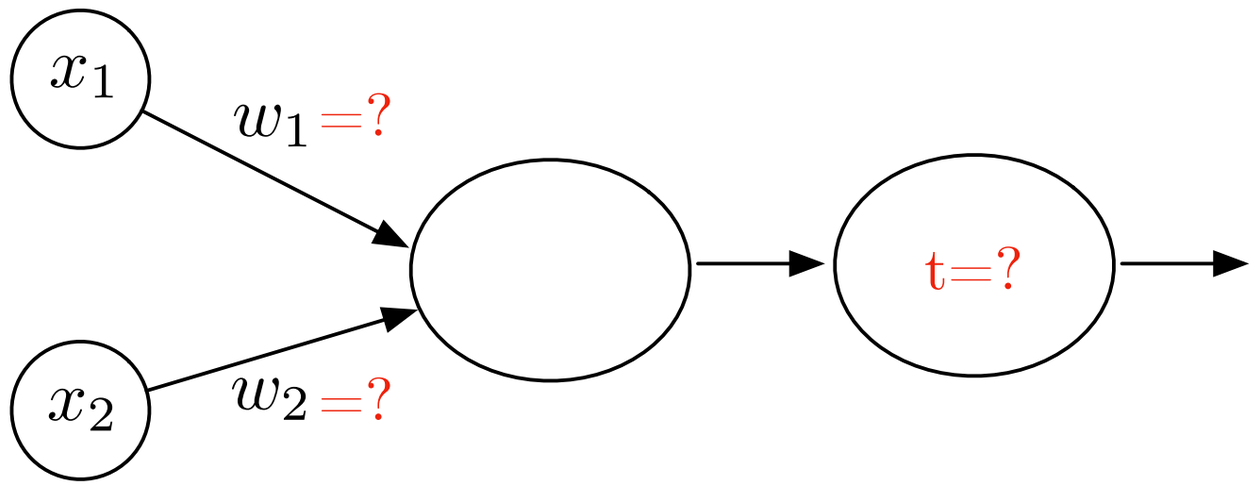
x_1	Out
0	1
1	0



Logical XOR Gate

(Take-home exercise)

x_1	x_2	Out
0	0	0
0	1	1
1	0	1
1	1	0



Rosenblatt's Perceptron

A learning rule for the computational/mathematical neuron model

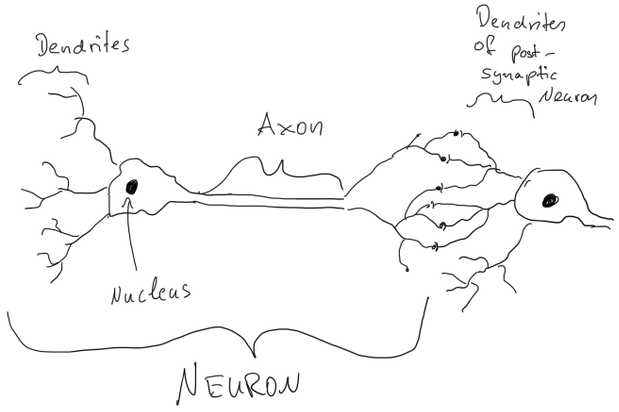
Rosenblatt, F. (1957). *The perceptron, a perceiving and recognizing automaton. Project Para*. Cornell Aeronautical Laboratory.

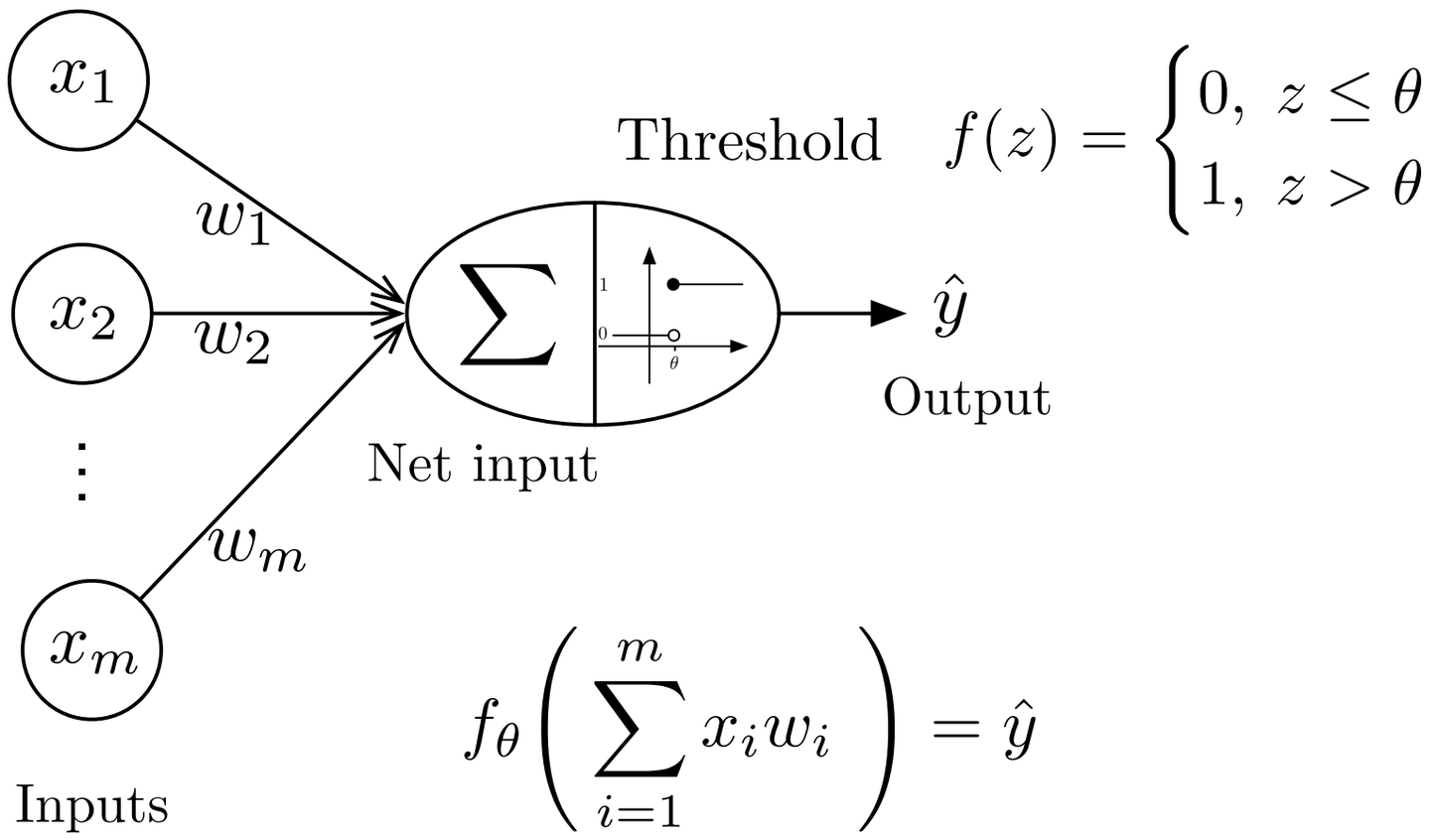
Source: http://www.enzyklopaedie-der-wirtschaftsinformatik.de/wi-enzyklopaedie/Members/wilex4/Rosen-2.jpg

Perceptron Variants

Note that Rosenblatt (and later others) proposed many variants of the Perceptron model and learning rule.
We discuss a "basic" version;
let's say, "Perceptron" := "a classic Rosenblatt Perceptron"

A Computational Model of a Biological Neuron





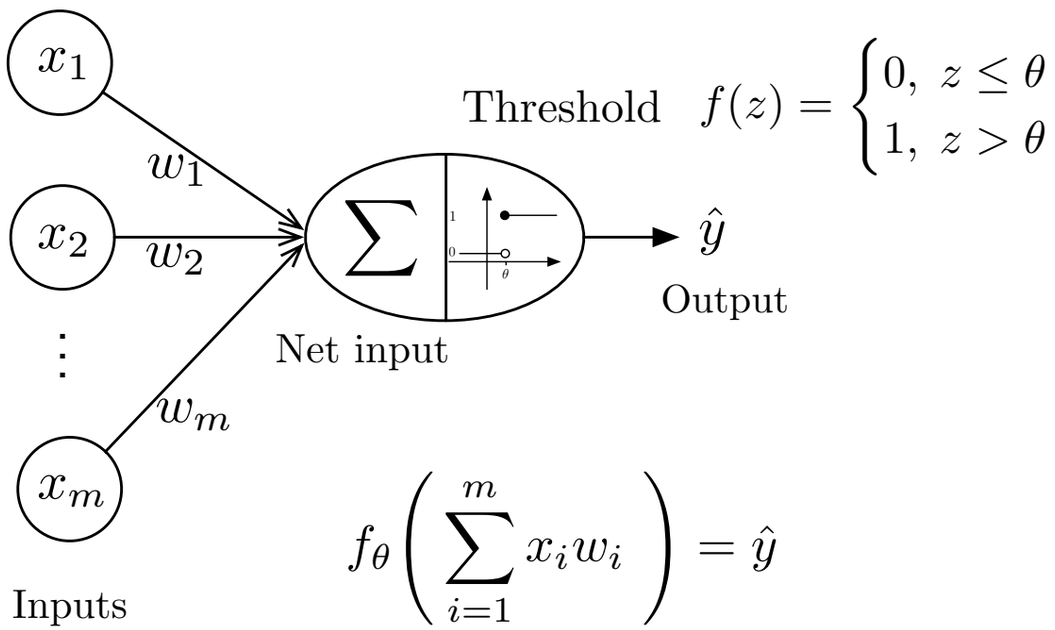
Terminology

General (logistic regression, multilayer nets, ...):

- Net input = weighted inputs
- Activations = activation function(net input)
- Label output = threshold(activations of last layer)

Special cases:

- In perceptron: activation function = threshold function
- In linear regression: activation = net input = output



Perceptron Output

$$\hat{y} = \begin{cases} 0, & z \leq \theta \\ 1, & z > \theta \end{cases}$$

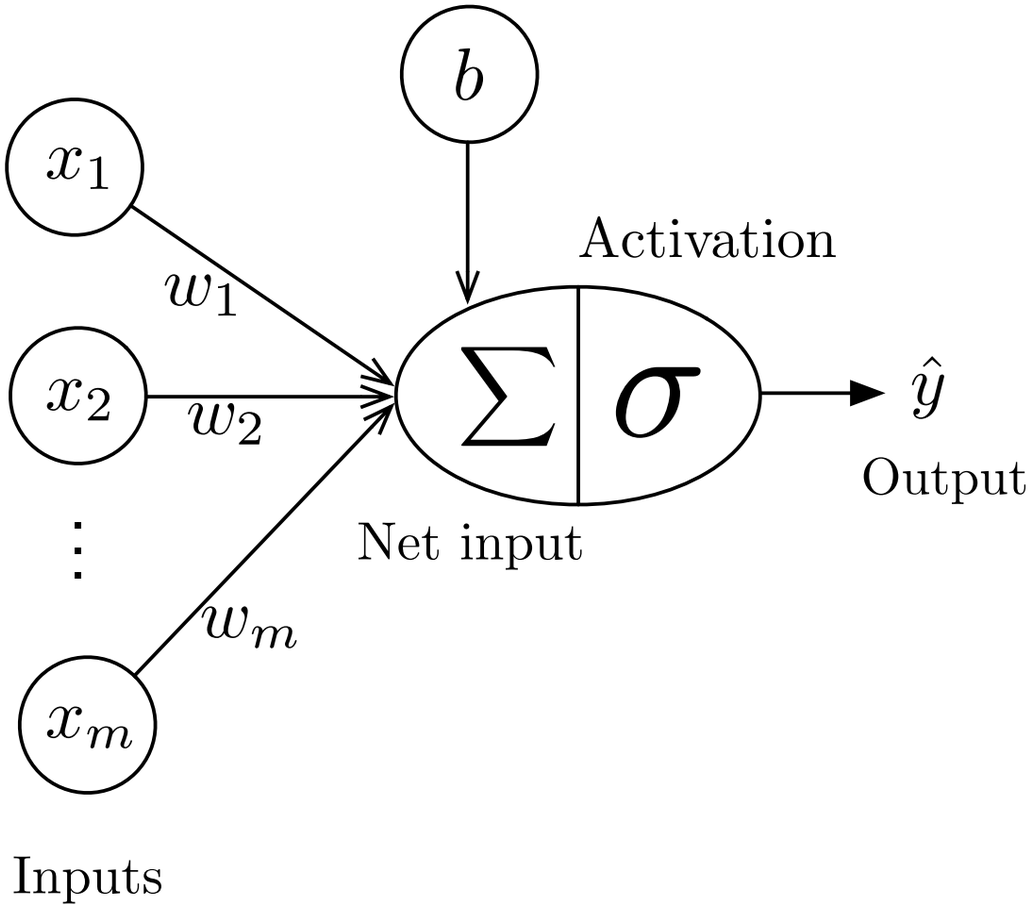
More convenient to re-arrange:

$$\hat{y} = \begin{cases} 0, & z - \theta \leq 0 \\ 1, & z - \theta > 0 \end{cases}$$

$-\theta$ = "bias"

General Notation for Single-Layer Neural Networks

- Common notation (in most modern texts): define the bias unit separately
- However, often inconvenient for mathematical notation



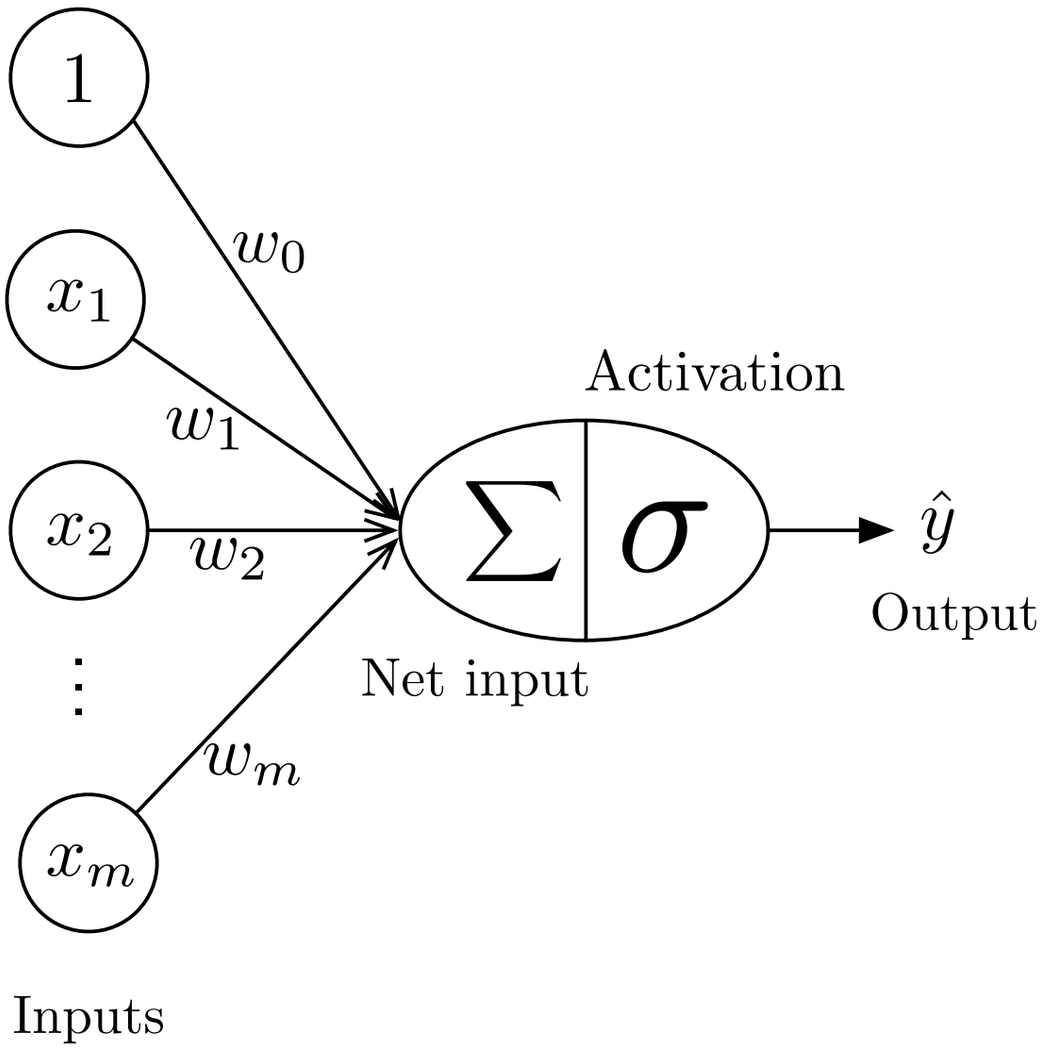
$$\sigma\left(\sum_{i=1}^{m} x_i w_i + b\right) = \sigma\left(\mathbf{x}^T\mathbf{w} + b\right) = \hat{y}$$

$$\sigma(z) = \begin{cases} 0, & z \leq 0 \\ 1, & z > 0 \end{cases}$$

$$b = -\theta$$

General Notation for Single-Layer Neural Networks

- Often more convenient notation: define bias unit as w_0 and prepend a 1 to <u>each</u> input vector as an additional "feature" value
- Modifying input vectors is more inconvenient/inefficient coding-wise, though

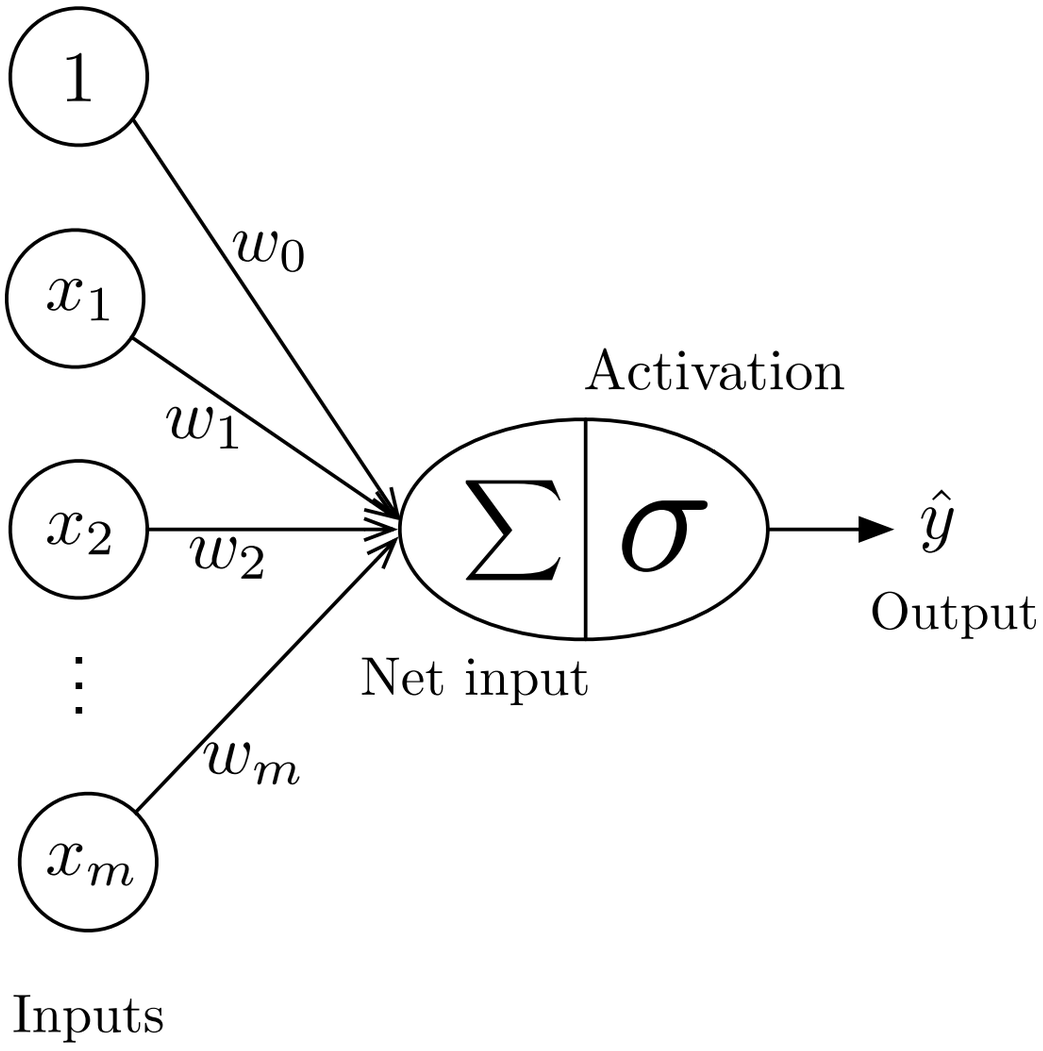


$$\sigma\left(\sum_{i=0}^{m} x_i w_i\right) = \sigma\left(\mathbf{x}^T \mathbf{w}\right) = \hat{y}$$

$$\sigma(z) = \begin{cases} 0, & z \leq 0 \\ 1, & z > 0 \end{cases}$$

$$w_0 = -\theta$$

General Notation for Single-Layer Neural Networks



Vector dot product

$$\sigma\left(\sum_{i=0}^{m} x_0 w_0\right) = \sigma\left(\mathbf{x}^T \mathbf{w}\right) = \hat{y}$$

$$\sigma(z) = \begin{cases} 0, & z - \theta \leq 0 \\ 1, & z - \theta > 0 \end{cases}$$

$$w_0 = -\theta$$

Interlude: Vectorization

In [1]:

```
x0, x1, x2 = 1., 2., 3.
bias, w1, w2 = 0.1, 0.3, 0.5

x = [x0, x1, x2]
w = [bias, w1, w2]
```

A simple for-loop:

In [2]:

```
z = 0.
for i in range(len(x)):
    z += x[i] * w[i]

print(z)
```

```
2.2
```

Interlude: Vectorization

A simple for-loop:

In [2]:

```
z = 0.
for i in range(len(x)):
    z += x[i] * w[i]

print(z)
```

```
2.2
```

A little bit better, list comprehensions:

In [3]:

```
z = sum(x_i*w_i for x_i, w_i in zip(x, w))
print(z)
```

```
2.2
```

Interlude: Vectorization

list comprehensions (still sequential):

In [3]:

```
z = sum(x_i*w_i for x_i, w_i in zip(x, w))
print(z)
```

```
2.2
```

A vectorized implementation:

In [4]:

```
import numpy as np

x_vec, w_vec = np.array(x), np.array(w)

z = (x_vec.transpose()).dot(w_vec)
print(z)

z = x_vec.dot(w_vec)
print(z)
```

```
2.2
2.2
```

Interlude: Vectorization

In [5]:
```
def forloop(x, w):
    z = 0.
    for i in range(len(x)):
        z += x[i] * w[i]
    return z


def listcomprehension(x, w):
    return sum(x_i*w_i for x_i, w_i in zip(x, w))


def vectorized(x, w):
    return x_vec.dot(w_vec)


x, w = np.random.rand(100000), np.random.rand(100000)
```

In [6]:
```
%timeit -r 100 -n 10 forloop(x, w)
```

```
38.9 ms ± 1.32 ms per loop (mean ± std. dev. of 100 r
uns, 10 loops each)
```

In [7]:
```
%timeit -r 100 -n 10 listcomprehension(x, w)
```

```
29.7 ms ± 842 µs per loop (mean ± std. dev. of 100 ru
ns, 10 loops each)
```

In [8]:
```
%timeit -r 100 -n 10 vectorized(x_vec, w_vec)
```

```
46.8 µs ± 8.07 µs per loop (mean ± std. dev. of 100 r
uns, 10 loops each)
```

Interlude: Connections and Parallel Computation

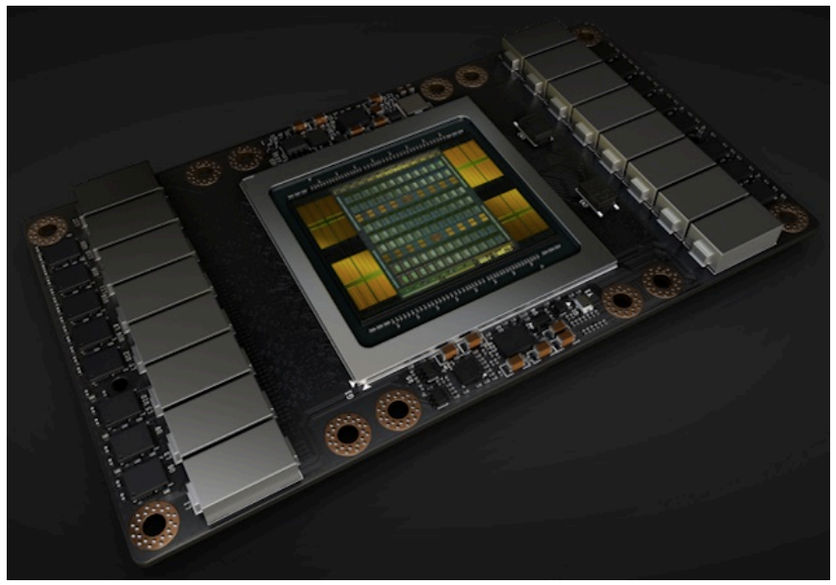


Image Source: https://fossbytes.com/wp-content/uploads/2017/05/nvidia-volta-v100-gpu.jpg

NVIDIA Volta with approx. 2.1 x 10^{10} transistors

approx. only 10 connections per transistor

Image Source: https://timedotcom.files.wordpress.com/2014/05/brain.jpg?w=1100&quality=85

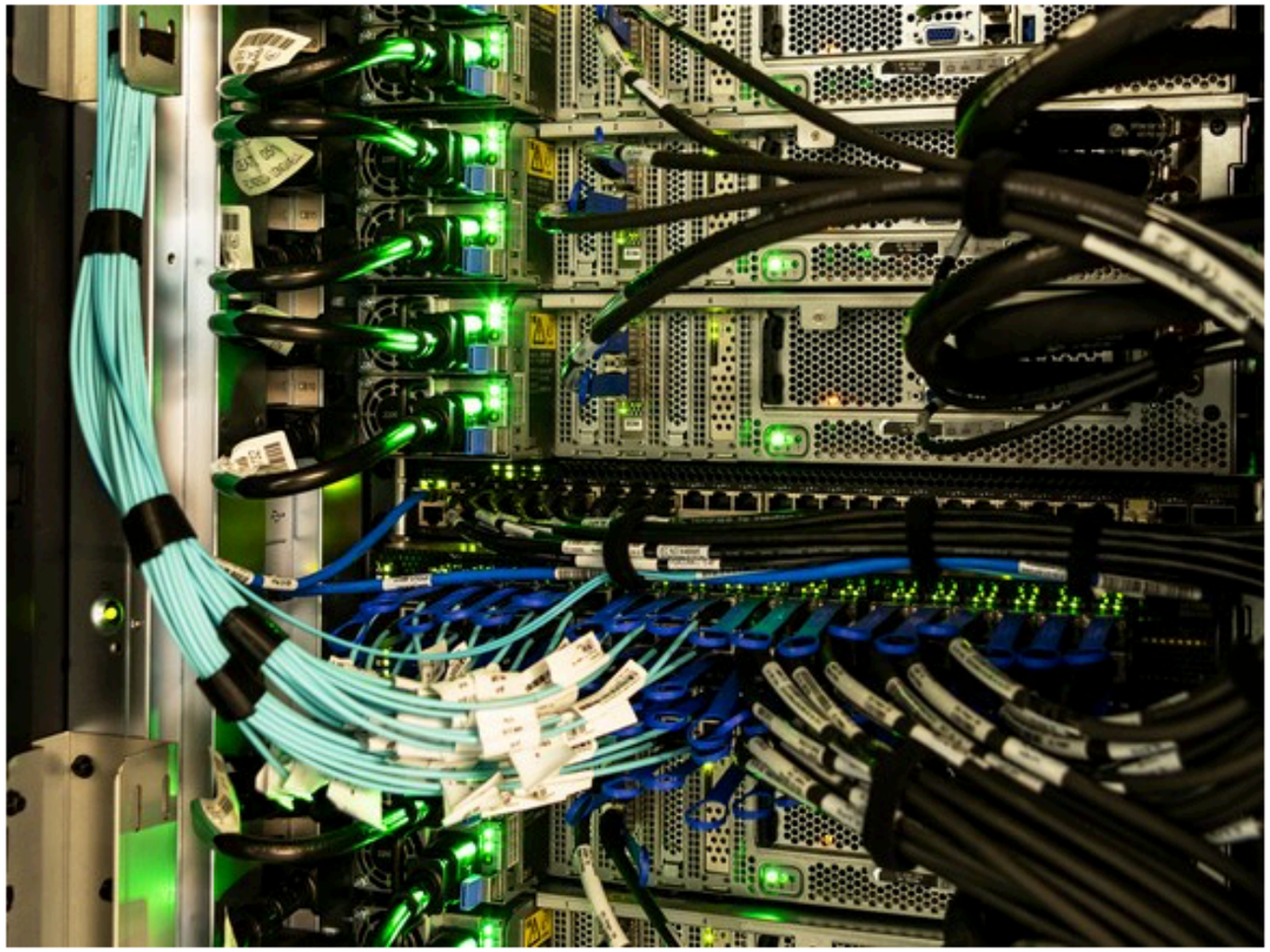
Brain with 1.6 x 10^{10} neurons

10^4 -10^5 connections per neuron

approx. 10^{15} connections in total

TOM SIMONITE BUSINESS 01.31.19 05:58 PM

THE WORLD'S FASTEST SUPERCOMPUTER BREAKS AN AI RECORD



Oak Ridge National Lab's Summit supercomputer became the world's most powerful in 2018, reclaiming that title from China for the first time in five years.

CARLOS JONES/OAK RIDGE NATIONAL LAB

Announced yesterday

https://www.wired.com/story/worlds-fastest-supercomputer-breaks-ai-record/

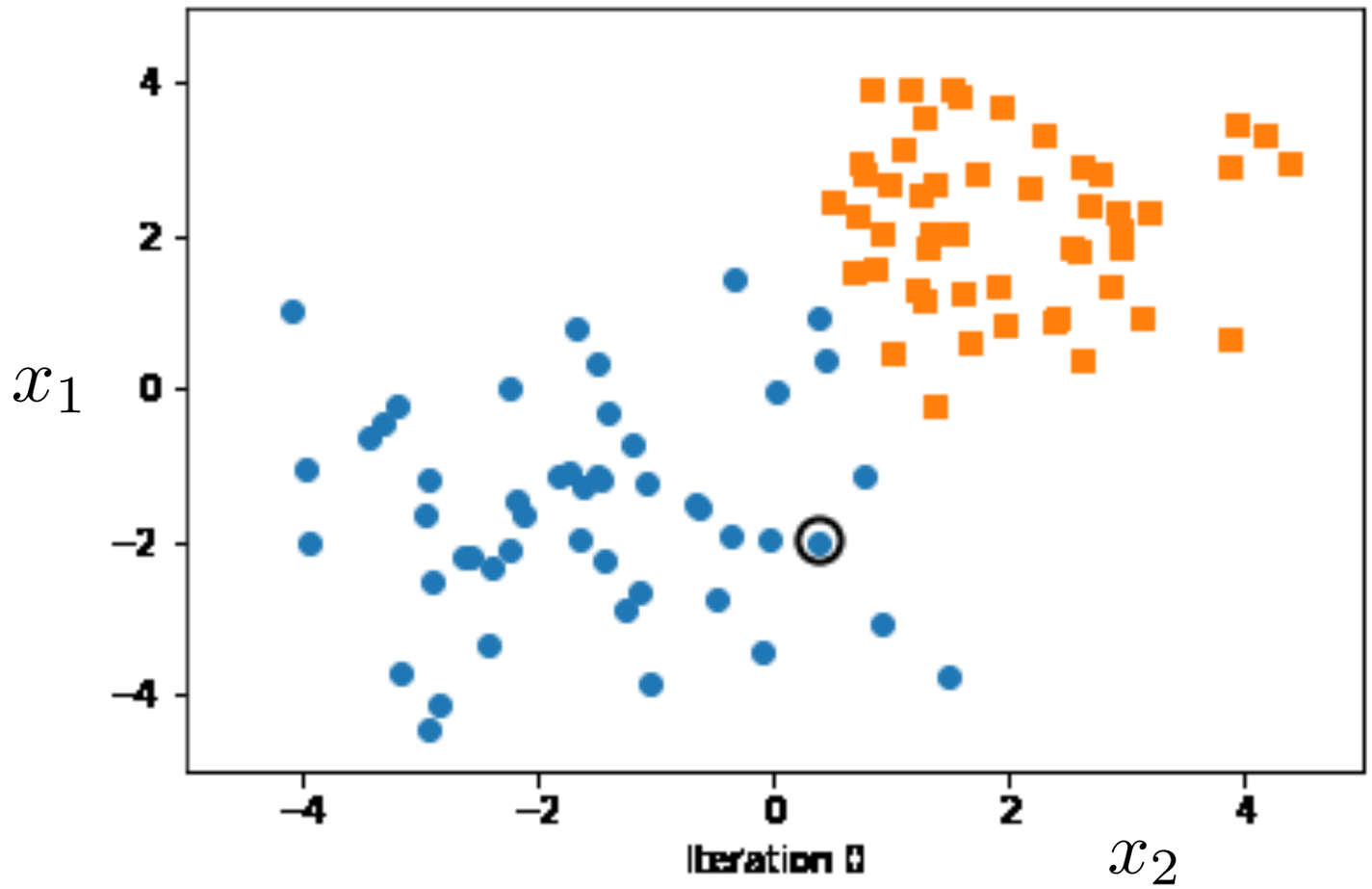
> 27,000 GPUs
"billion billion" operations per second (exaflop)

"Deep learning has never been scaled to such levels of performance before," says Prabhat (research group leader at Berkeley National Lab)

Application: Weather patterns (three-hour forecasts)

Perceptron Learning Rule

Assume binary classification task, Perceptron finds decision boundary if classes are separable

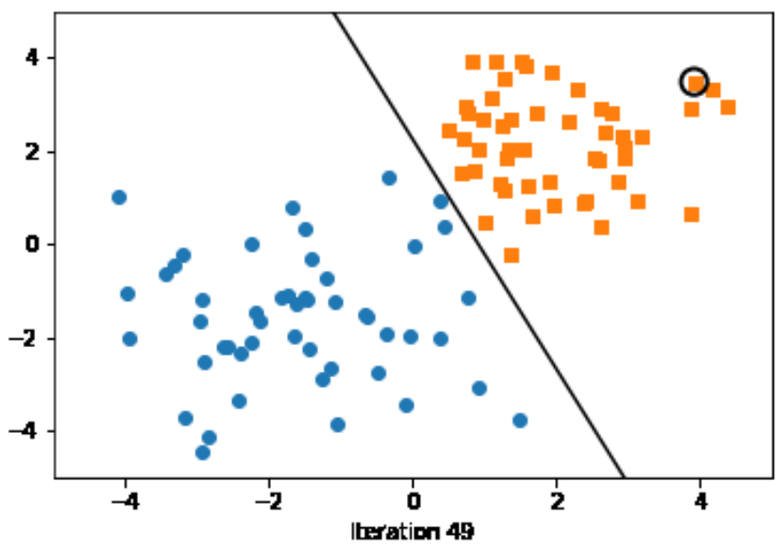


[animated GIF]

Code at https://github.com/rasbt/stat479-deep-learning-ss19/tree/master/L03_perceptron/code/perceptron-animation.ipynb if you want to play with it.

The Perceptron Learning Algorithm

- If correct: Do nothing if the prediction if output is equal to the target
- If incorrect, scenario a):
  If output is 0 and target is 1, add input vector to weight vector
- If incorrect, scenario b):
  If output is 1 and target is 0, subtract input vector from weight vector



Guaranteed to converge if a solution exists
(more about that later...)

The Perceptron Learning Algorithm

Let

$$\mathcal{D} = (\langle \mathbf{x}^{[1]}, y^{[1]} \rangle, \langle \mathbf{x}^{[2]}, y^{[2]} \rangle, ..., \langle \mathbf{x}^{[n]}, y^{[n]} \rangle) \in (\mathbb{R}^m \times \{0, 1\})^n$$

1. Initialize $\mathbf{w} := 0^m$ (assume notation where weight incl. bias)
2. For every training epoch:
   A. For every $\langle \mathbf{x}^{[i]}, y^{[i]} \rangle \in \mathcal{D}$:
      (a) $\hat{y}^{[i]} := \sigma\left(\mathbf{x}^{[i]T}\mathbf{w}\right)$
      (b) $\text{err} := (y^{[i]} - \hat{y}^{[i]})$
      (c) $\mathbf{w} := \mathbf{w} + err \times \mathbf{x}^{[i]}$

Perceptron Coding Example

https://github.com/rasbt/stat479-deep-learning-ss19/tree/master/L03_perceptron/code/perceptron-numpy.ipynb

Perceptron Convergence Theorem

Let

$$\mathcal{D} = (\langle \mathbf{x}^{[1]}, y^{[1]} \rangle, \langle \mathbf{x}^{[2]}, y^{[2]} \rangle, ..., \langle \mathbf{x}^{[n]}, y^{[n]} \rangle) \in (\mathbb{R}^m \times \{0, 1\})^n$$

$$\forall y^{[i]} \in \mathcal{D}_1 : y^{[i]} = 1$$
$$\forall y^{[i]} \in \mathcal{D}_2 : y^{[i]} = 0$$

and $\mathcal{D}_1 \cup \mathcal{D}_2 = \mathcal{D}$

Assume the input vectors come from two linearly separable classes such that a feasible weight vector $\mathbf{w}^*$ exists.

The perceptron learning algorithm is guaranteed to converge to a weight vector in the feasible region in a finite number of iterations such that

$$\forall \mathbf{x}^{[i]} \in \mathcal{D}_1 : \mathbf{w}^T \mathbf{x}^{[i]} > 0$$
$$\forall \mathbf{x}^{[i]} \in \mathcal{D}_2 : \mathbf{w}^T \mathbf{x}^{[i]} \leq 0$$

Perceptron Convergence Theorem -- Proof

Let us slightly rewrite the update rule (upon misclassification) for convenience when we construct the proof:

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \;\; \text{if} \;\; (\mathbf{w}^{[i]})^T\mathbf{x}^{[i]} \leq 0 \; , \mathbf{x}^{[i]} \in \mathcal{D}_1$$

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} - \mathbf{x}^{[i]} \;\; \text{if} \;\; (\mathbf{w}^{[i]})^T\mathbf{x}^{[i]} > 0 \; , \mathbf{x}^{[i]} \in \mathcal{D}_2$$

Here, I mean the weight vector that the next training example will use

Perceptron Convergence Theorem -- Proof

From the previous slide:

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \;\; \text{if} \;\; (\mathbf{w}^{[i]})^T \mathbf{x}^{[i]} \leq 0 \; , \mathbf{x}^{[i]} \in \mathcal{D}_1$$

We can rewrite this as follows:

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[0]} + \mathbf{x}^{[1]} + ... + \mathbf{x}^{[i]}$$

Also, we can drop this term if we initialize the weight vector as 0^m

$$\mathbf{w}^{[i+1]} = \mathbf{x}^{[1]} + ... + \mathbf{x}^{[i]}$$

Perceptron Convergence Theorem -- Proof

From the previous slide, the update rule:

$$\mathbf{w}^{[i+1]} = \mathbf{x}^{[1]} + ... + \mathbf{x}^{[i]}$$

Let's multiply both sides by $\mathbf{w}^*$:

$$(\mathbf{w}^*)^T\mathbf{w}^{[i+1]} = (\mathbf{w}^*)^T\mathbf{x}^{[1]} + ... + (\mathbf{w}^*)^T\mathbf{x}^{[i]}$$

All these terms are > 0, because remember that we have

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \;\text{ if }\; (\mathbf{w}^{[i]})^T\mathbf{x}^{[i]} \leq 0 \;, \mathbf{x}^{[i]} \in \mathcal{D}_1$$

so the updates are all to make the net inputs more positive

Now, let $\alpha = \min_{x^{[j]}} (\mathbf{w}^*)^T\mathbf{x}^{[j]},\; j = 1, ..., i$

then $(\mathbf{w}^*)^T\mathbf{w}^{[i+1]} \geq \alpha i$

Perceptron Convergence Theorem -- Proof

From the previous slide, we had the inequality:

$$\underbrace{(\mathbf{w}^*)^T\mathbf{w}}^{[i+1]} \geq \alpha i$$

Using the Cauchy-Schwarz inequality, we can then say

$$||\mathbf{w}^*||^2 \cdot ||\mathbf{w}^{[i+1]}||^2 \geq \left((\mathbf{w}^*)^T\mathbf{w}^{[i+1]}\right)^2$$

as well as

$$||\mathbf{w}^*||^2 \cdot ||\mathbf{w}^{[i+1]}||^2 \geq (\alpha i)^2$$

So, we can finally define the lower bound of the size of the weights

$$||\mathbf{w}^{[i+1]}||^2 \geq \frac{\alpha^2 i^2}{||\mathbf{w}^*||^2}$$

Perceptron Convergence Theorem -- Proof

Now that we defined the lower bound of the size of the weights, let us get the upper bound.

For that, let's go back to the update rule

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \quad \text{if} \quad (\mathbf{w}^{[i]})^T \mathbf{x}^{[i]} \leq 0 \, , \mathbf{x}^{[i]} \in \mathcal{D}_1$$

and apply the squared L2 norm on both sides

$$||\mathbf{w}^{[i+1]}||^2 = ||\mathbf{w}^{[i]} + \mathbf{x}^{[i]}||^2$$

$$= ||\mathbf{w}^{[i]}||^2 + 2(\mathbf{x}^{[i]})^T \mathbf{w}^{[i]} + ||\mathbf{x}^{[i]}||^2$$

Perceptron Convergence Theorem -- Proof

Now that we defined the lower bound of the size of the weights, let us get the upper bound.

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$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \;\; \text{if} \;\; (\mathbf{w}^{[i]})^T\mathbf{x}^{[i]} \leq 0 \; , \mathbf{x}^{[i]} \in \mathcal{D}_1$$

and apply the squared L2 norm on both sides

$$||\mathbf{w}^{[i+1]}||^2 = ||\mathbf{w}^{[i]} + \mathbf{x}^{[i]}||^2$$
$$= ||\mathbf{w}^{[i]}||^2 + 2(\mathbf{x}^{[i]})^T\mathbf{w}^{[i]} + ||\mathbf{x}^{[i]}||^2$$

Leads to

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[i]}||^2 + ||\mathbf{x}^{[i]}||^2$$

Perceptron Convergence Theorem -- Proof

Now that we defined the lower bound of the size of the weights, let us get the upper bound.

For that, let's go back to the update rule

$$\mathbf{w}^{[i+1]} = \mathbf{w}^{[i]} + \mathbf{x}^{[i]} \text{ if } \underbrace{(\mathbf{w}^{[i]})^T \mathbf{x}^{[i]} \leq 0}_{\text{implies}} , \mathbf{x}^{[i]} \in \mathcal{D}_1$$

and apply the squared L2 norm on both sides

$$||\mathbf{w}^{[i+1]}||^2 = ||\mathbf{w}^{[i]} + \mathbf{x}^{[i]}||^2$$
$$= ||\mathbf{w}^{[i]}||^2 + \underbrace{2(\mathbf{x}^{[i]})^T \mathbf{w}^{[i]}}_{\leq 0} + ||\mathbf{x}^{[i]}||^2$$

Thus

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[i]}||^2 + ||\mathbf{x}^{[i]}||^2$$

Perceptron Convergence Theorem -- Proof

Now, we simply expand:

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[i]}||^2 + ||\mathbf{x}^{[i]}||^2$$

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[i-1]}||^2 + ||\mathbf{x}^{[i-1]}||^2 + ||\mathbf{x}^{[i]}||^2$$

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[i-2]}||^2 + ||\mathbf{x}^{[i-2]}||^2 + ||\mathbf{x}^{[i-1]}||^2 + ||\mathbf{x}^{[i]}||^2$$

$$\dots$$

$$||\mathbf{w}^{[i+1]}||^2 \leq ||\mathbf{w}^{[1]}||^2 + \sum_{j=1}^{i} ||\mathbf{x}^{[j]}||^2$$

$$||\mathbf{w}^{[i+1]}||^2 \leq \sum_{j=1}^{i} ||\mathbf{x}^{[j]}||^2$$

Perceptron Convergence Theorem -- Proof

From $||\mathbf{w}^{[i+1]}||^2 \leq \sum_{j=1}^{i} ||\mathbf{x}^{[j]}||^2$ we can finally get the upper bound.

Let $\beta = \max ||x^{[j]}||^2$

then $||\mathbf{w}^{[i+1]}||^2 \leq \beta i$

Perceptron Convergence Theorem -- Proof

lower bound

$$||\mathbf{w}^{[i+1]}||^2 \geq \frac{\alpha^2 i^2}{||\mathbf{w}^*||^2}$$

upper bound

$$||\mathbf{w}^{[i+1]}||^2 \leq \beta i$$

combined

$$\beta_i \geq ||\mathbf{w}^{[i+1]}||^2 \geq \frac{\alpha^2 i^2}{||\mathbf{w}^*||^2}$$

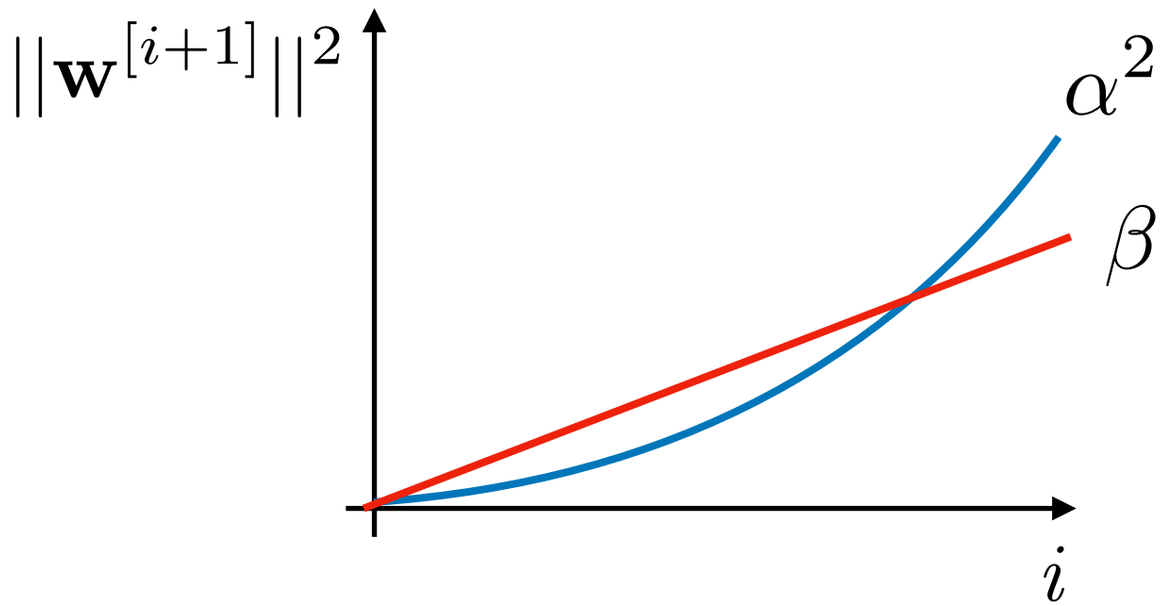
$$i \leq \frac{\beta ||\mathbf{w}^*||^2}{\alpha^2}$$

Since the number of iterations *i* has an upper bound,
we can conclude that the weights only change a finite number of times and will converge if the classes are linearly separable.

□

Perceptron Convergence Theorem -- Proof

$$\beta i \geq ||\mathbf{w}^{[i+1]}||^2 \geq \alpha^2 i^2$$



Perceptron Convergence Theorem -- Proof

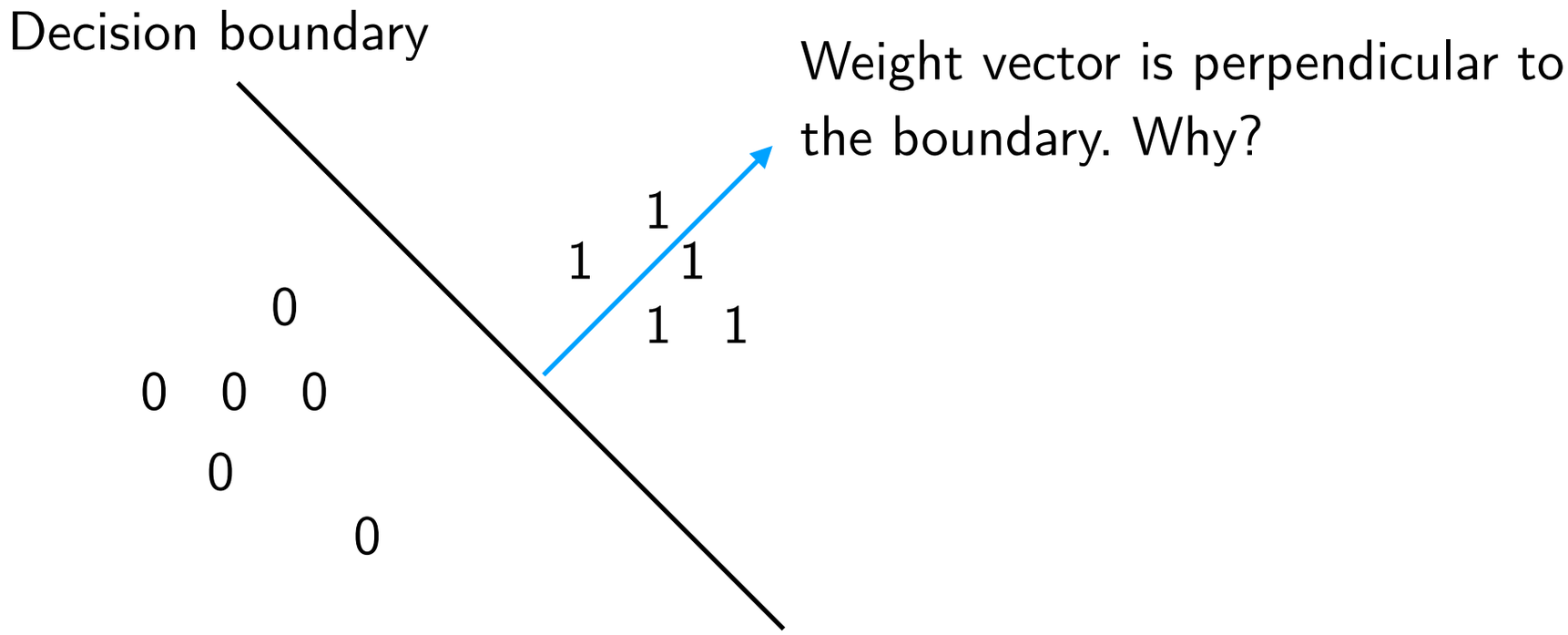
In the convergence theorem, we can assume that $||\mathbf{w}^*|| = 1$

(so you may remove it from all equations)

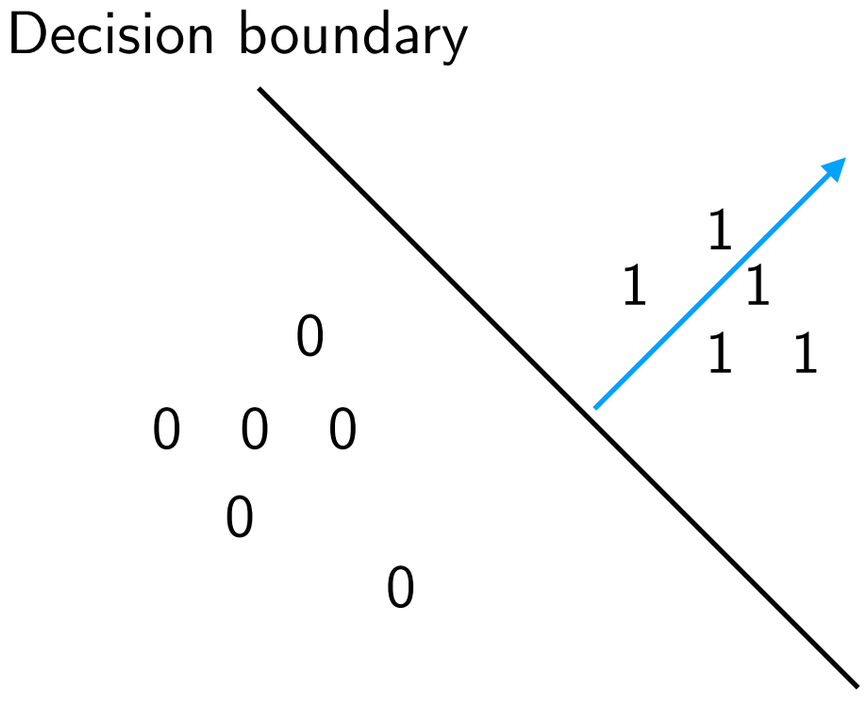
$$\beta_i \geq ||\mathbf{w}^{[i+1]}||^2 \geq \frac{\alpha^2 i^2}{||\mathbf{w}^*||^2} \quad \Leftrightarrow \quad \beta_i \geq ||\mathbf{w}^{[i+1]}||^2 \geq \alpha^2 i^2$$

$$i \leq \frac{\beta||\mathbf{w}^*||^2}{\alpha^2} \quad \Leftrightarrow \quad i \leq \frac{\beta}{\alpha^2}$$

Geometric Intuition



Geometric Intuition



Weight vector is perpendicular to the boundary. Why?

Remember,

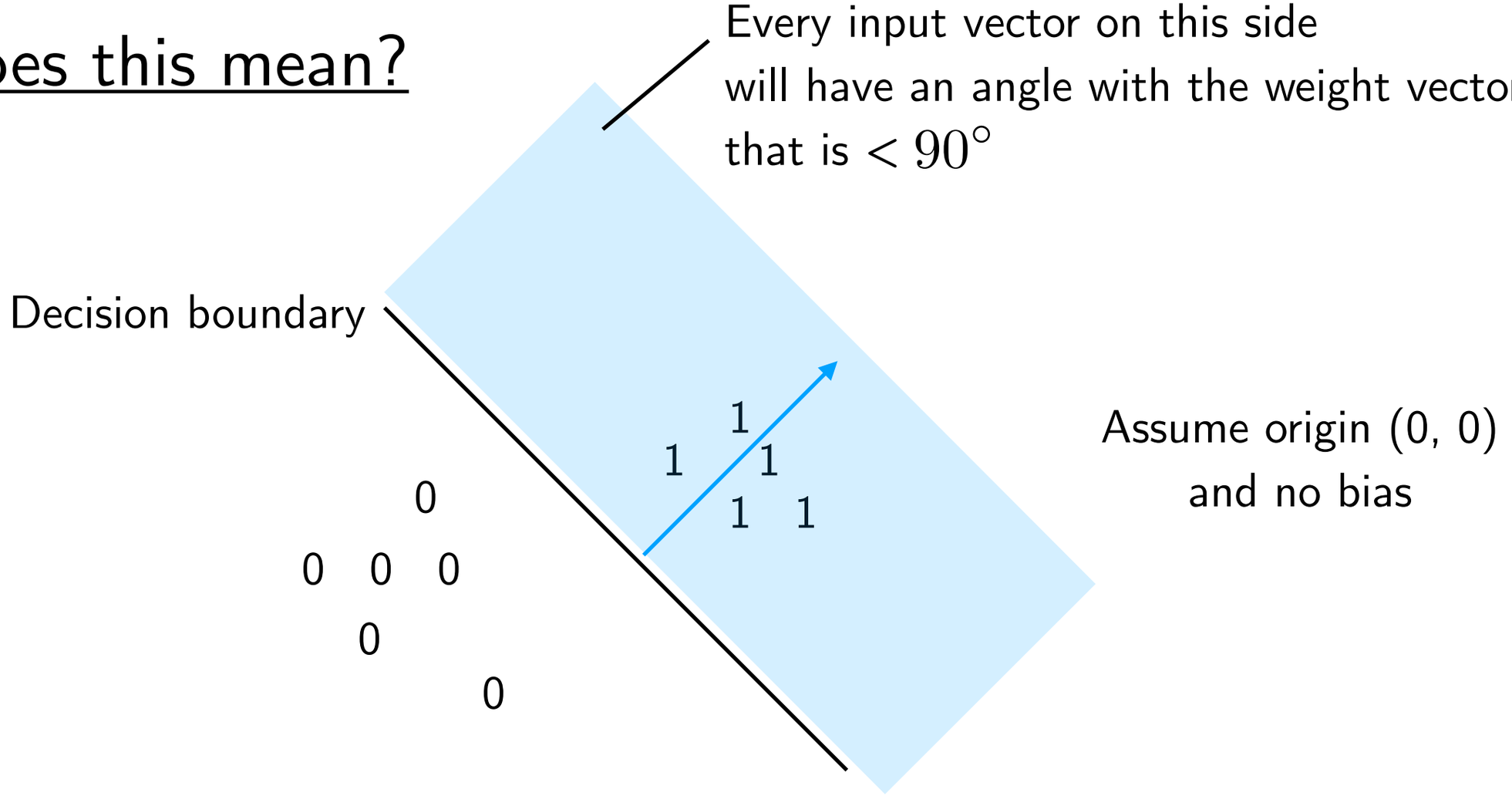
$$\hat{y} = \begin{cases} 0, & \mathbf{w}^T\mathbf{x} \leq 0 \\ 1, & \mathbf{w}^T\mathbf{x} > 0 \end{cases}$$

$$\mathbf{w}^T\mathbf{x} = ||\mathbf{w}|| \cdot ||\mathbf{x}|| \cdot \underbrace{\cos(\theta)}$$

So this needs to be 0 at the boundary, and it is zero at 90°

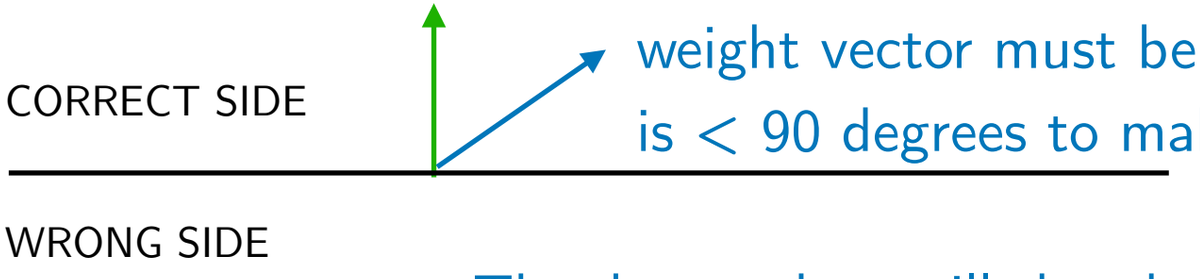
Geometric Intuition

What else does this mean?



So, we could scale the weights and/or inputs by an arbitrary factor and still get the same classification results
(but large inputs will take much longer to converge if you check the bounds we defined previously ...)

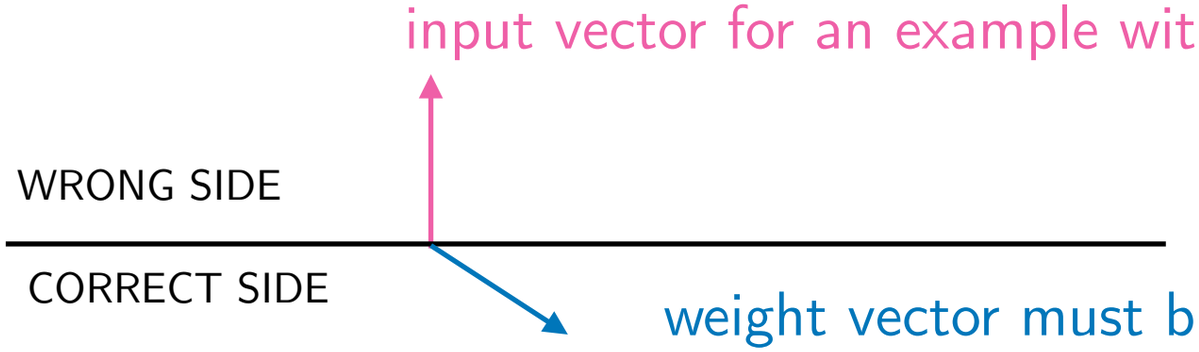
Geometric Intuition



The dot product will then be positive, i.e., > 0, since

$$\mathbf{w}^T\mathbf{x} = ||\mathbf{w}|| \cdot ||\mathbf{x}|| \cdot \cos(\theta)$$

Geometric Intuition

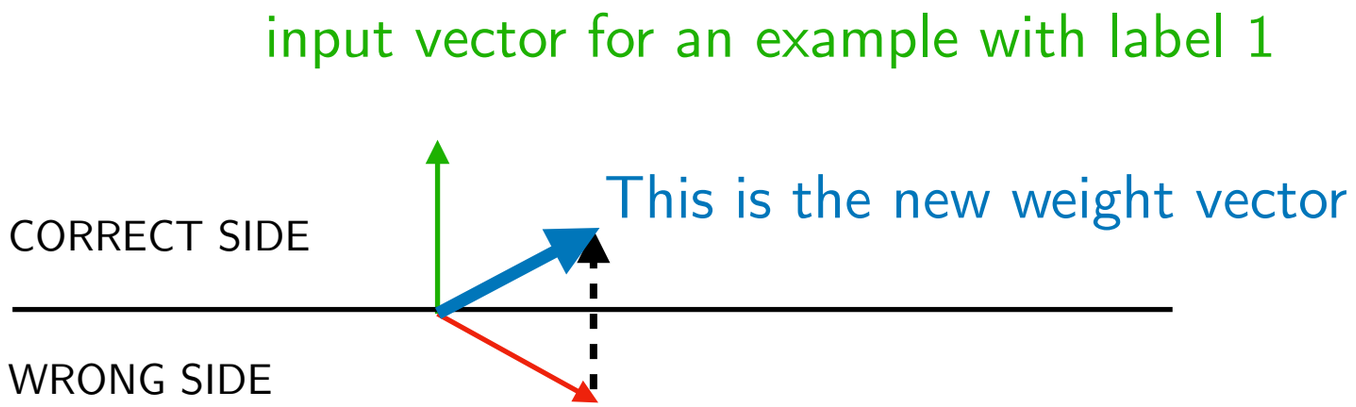


weight vector must be somewhere such that the angle is $\geq$ 90 degrees to make a correct prediction

The dot product will then $\leq$ 0, since

$$\mathbf{w}^T\mathbf{x} = ||\mathbf{w}|| \cdot ||\mathbf{x}|| \cdot \cos(\theta)$$

Geometric Intuition



For this weight vector, we make a wrong prediction; hence, we update

Perceptron Conclusions

The (classic) Perceptron has many problems
(as discussed in the previous lecture)

- Linear classifier, no non-linear boundaries possible
- Binary classifier, cannot solve XOR problems, for example
- Does not converge if classes are not linearly separable
- Many "optimal" solutions in terms of 0/1 loss on the training data, most will not be optimal in terms of generalization performance

Perceptron Fun Fact

[...] Where a perceptron had been trained to distinguish between - this was for military purposes - it was looking at a scene of a forest in which there were camouflaged tanks in one picture and no camouflaged tanks in the other. And the perceptron - after a little training - made a 100% correct distinction between these two different sets of photographs. Then they were embarrassed a few hours later to discover that the two rolls of film had been developed differently. And so these pictures were just a little darker than all of these pictures and the perceptron was just measuring the total amount of light in the scene. But it was very clever of the perceptron to find some way of making the distinction.

-- Marvin Minsky, Famous AI researcher, Author of the famous "Perceptrons" book

Source: https://www.webofstories.com/play/marvin.minsky/122

An Analogy to Future Lectures ...

We can say the perceptron optimizes a loss function analogous to the squared error in least-squares regression, except that we have targets and outputs:

$$\mathcal{L}(\mathbf{w}, b) = \sum_i \frac{1}{2}(\hat{y}^{[i]} - y^{[i]})^2$$

where $\hat{y} = \sigma(\mathbf{w}^T\mathbf{x} + b) = \begin{cases} 0, & \mathbf{w}^T\mathbf{x} + b \leq 0 \\ 1, & \mathbf{w}^T\mathbf{x} + b > 0 \end{cases}$

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$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial w_j} &= \frac{\partial}{\partial w_j} \sum_i (\hat{y}^{[i]} - y^{[i]})^2 \\
&= \frac{\partial}{\partial w_j} \sum_i (\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]})^2 \\
&= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \frac{\partial}{\partial w_j} (\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \\
&= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \sigma'(\mathbf{w}^T\mathbf{x}^{[i]}) \frac{\partial}{\partial w_j} \mathbf{w}^T\mathbf{x}^{[i]} \\
&= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \sigma'(\mathbf{w}^T\mathbf{x}^{[i]}) x_j^{[i]}
\end{aligned}$$

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$$= \frac{\partial}{\partial w_j} \sum_i (\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]})^2$$

$$= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \frac{\partial}{\partial w_j} (\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]})$$

$$= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \sigma'(\mathbf{w}^T\mathbf{x}^{[i]}) \frac{\partial}{\partial w_j} \mathbf{w}^T\mathbf{x}^{[i]}$$

$$= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \sigma'(\mathbf{w}^T\mathbf{x}^{[i]}) x_j^{[i]}$$

not differentiable!

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&= \sum_i 2(\sigma(\mathbf{w}^T\mathbf{x}^{[i]}) - y^{[i]}) \sigma'(\mathbf{w}^T\mathbf{x}^{[i]}) x_j^{[i]}
\end{aligned}$$

not differentiable!

However, perceptron does something very similar to stochastic gradient descent:

(not a real derivative)

$$\frac{\partial \mathcal{L}}{\partial w_j} = (y^{[i]} - \hat{y}^{[i]}) x_j$$

$$w_j := w_j + \frac{\partial \mathcal{L}}{\partial w_j}$$

On Deep Learning vs How the Brain Works

MARTIN FORD: You gave an interview toward the end of 2017 where you said that you were suspicious of the backpropagation algorithm and that it needed to be thrown out and we needed to start from scratch.1 That created a lot of disturbance, so I wanted to ask what you meant by that?

GEOFFREY HINTON: The problem was that the context of the conversation wasn't properly reported. I was talking about trying to understand the brain, and I was raising the issue that backpropagation may not be the right way to understand the brain. We don't know for sure, but there are some reasons now for believing that the brain might not use backpropagation. I said that if the brain doesn't use backpropagation, then whatever the brain is using would be an interesting candidate for artificial systems. I didn't at all mean that we should throw out backpropagation. Backpropagation is the mainstay of all the deep learning that works, and I don't think we should get rid of it.

---

1 See: https://www.axios.com/artificial-intelligence-pioneer-says-we-need-to-start-over-1513305524-f619efbd-9db0-4947-a9b2-7a4c310a28fe.html

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(Excerpt from "Architects of Intelligence")

Homework Assignment 1

Your task:

Based on the NumPy example,
implement Perceptron code into pure Python
(= no NumPy, no PyTorch, etc.)

1) Download code from:

https://github.com/rasbt/stat479-deep-learning-ss19/tree/master/hw1

2) Submit solution via Canvas for grading